

WORKSHEET 01

CUQ

1. $\frac{d}{dx}(x^{2025}) =$

1) $2025.x^{2023}$

2) $2025.x^{2024}$

3) $2025x^{2010}$

4) $2005x^{2024}$

2. $\frac{d}{dx}\left(x^{\frac{2}{3}}\right) =$

1) $\frac{2}{3}x^{-\frac{1}{3}}$

2) $\frac{1}{3}x^{-\frac{2}{3}}$

3) $\frac{2}{3x^{\frac{1}{3}}}$

4) Both 1 and 3

3. $\frac{d}{dx}\left(\sin\left(x - \frac{\pi}{8}\right)\right) =$

1) $\sin\left(x - \frac{\pi}{8}\right)$

2) $-\sin\left(x - \frac{\pi}{8}\right)$

3) $\cos\left(x - \frac{\pi}{8}\right)$

4) $-\cos\left(x - \frac{\pi}{8}\right)$

4. $\frac{d}{dx}(10^x + \log_e x)\Big|_{x=0} =$

1) $\log 10$

2) 5

3) 4

4) not defined

5. $\frac{d}{dx}(\tan 3x) =$

1) $\sec 3x \cdot \tan 3x$

2) $3 \sec 3x \cdot \tan 3x$

3) $\sec^2 3x$

4) $3 \sec^2 3x$

6. $\frac{d}{dx}(4\sqrt{x} - 2) =$

1) $\frac{2}{\sqrt{x}}$

2) $\frac{2}{x}$

3) $\frac{4}{\sqrt{x}}$

4) $\frac{4}{x}$

7. $\frac{d}{dx}(x^{10}) =$

1) $10x$

2) $10x^9$

3) x^9

4) $\frac{x^9}{10}$

8. $\frac{d}{dx}(e^{5x} + \sin 3x) =$

1) $e^{5x} + 3 \cos 3x$

2) $5e^{5x} + \cos 3x$

3) $5e^{5x} + 3 \cos 3x$

4) $5e^{5x} - 3 \cos 3x$

9. $\frac{d}{dx} \cos(5x + 3) =$

- 1) $5 \sin(5x + 3)$ 2) $-\sin(5x + 3)$ 3) $-5 \sin(x + 3)$ 4) $-5 \sin(5x + 3)$

10. $\frac{d}{dx} (\operatorname{cosec} 5x) =$

- 1) $-5 \operatorname{cosec} 5x \cot 5x$ 2) $5 \operatorname{cosec} 5x \cot 5x$ 3) $-\operatorname{cosec} 5x \cot 5x$ 4) $\operatorname{cosec} 5x \cot 5x$

JEE MAIN & ADVANCED

LEVEL - 1

Single Correct Choice Type:

1. $\frac{d}{dx} (x^5 - x^3 - e^{3x} + 6^{2x}) =$

- 1) $5x^4 - 3x^2 - e^{3x} + 6^{2x} \log 2$ 2) $5x^4 - 3x^2 - 3e^{3x} + 6^{2x} \log 6$
 3) $5x^4 - 3x^2 - 3e^{3x} + 2 \cdot 6^{2x} \log 6$ 4) $5x^4 - 3x^2 - 3e^{3x} + 6e^{2x} \log 6$

2. $\frac{d}{dx} (x^2 - 1)^3 =$

- 1) $6x(x^2 - 1)^2$ 2) $6x(x^2 - 1)$ 3) $6(x^2 - 1)^2$ 4) $6(x^2 - 1)$

3. $\frac{d}{dx} (\cos(\sin x^2)) =$

- 1) $-2 \sin(\sin x^2) \cos x^2$ 2) $-2 \cos(\cos x^2) \cdot \sin x^2$
 3) $-2x \sin(\sin x^2) \cos x^2$ 4) $-2x \cos(\cos x^2) \cdot \sin x^2$

4. If $\frac{d}{dx} (e^{kx} + \sin x) = \cos x + 5e^{5x}$, then $k =$

- 1) 5 2) -5 3) 1 4) 0

5. If $\frac{d}{dx} [\cos(\theta x)] = -2010 \sin(2010x)$, then $\theta =$

- 1) -2010 2) 2009 3) 2011 4) 2010

Comprehension Type:

Let $y = f(u)$ be a function in 'u' and $u = g(x)$ be a function in x

1. Then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ is known as chain rule

2. $\frac{d}{dx} (f(x) \pm g(x)) = \frac{d}{dx} f(x) \pm \frac{d}{dx} g(x)$ using this two rules compute the following derivatives for the given functions

6. $3x^2 - 5x + 3 =$

1) $6x - 5$

2) $3x + 5$

3) $6x + 5$

4) $3x - 5$

7. $e^x - \sec x - \cot x =$

1) $e^x - \tan^2 x + \cot x$

2) $e^x - \sec x \cdot \tan x - \cot^2 x$

3) $e^x - \sec x \cdot \tan x + \operatorname{cosec}^2 x$

4) $e^x - \tan^2 x + \cot^2 x$

8. $\left(x + \frac{1}{x}\right)^3 =$

1) $3x^2 - \frac{3}{x^4} + 3 - \frac{3}{x^2}$

2) $3x^2 + \frac{3}{x^4} - 3 + \frac{3}{x^2}$

3) $3x^2 + \frac{3}{x^4} - 3 - \frac{3}{x^2}$

4) $3x^4 + \frac{3}{x^4} - 3 - \frac{3}{x^2}$

Multi Correct Choice Type:

9. If $\frac{d}{dx}[f(x)] = 6x^5 + 4e^{4x}$, then $f(x) =$

1) $x^6 + e^{-4x}$

2) $x^6 + e^{4x}$

3) $x^{-6} + e^{4x}$

4) $6x + e^{4x}$

10. Among the following the correct statement is

1) $\frac{d}{dx}(\cos 5x) = -\cos 5x$

2) $\frac{d}{dx}(\tan 5x) = 5 \sec^2 5x$

3) $\frac{d}{dx}(\sec 5x) = 5 \sec 5x \tan 5x$

4) $\frac{d}{dx}(2010) = 1$

Statement Type:

11. **Statement-I:** $\frac{d}{dx}(\operatorname{cosec} ax) = -a \operatorname{cosec} ax \cdot \cot ax$

Statement-II: The derivative of $(\sin x)^{\cos x}$ ($0 < x < \pi$) w.r.t x is

$(\sin x)^{\cos x} [\cos x \cot x - \sin x \log \sin x]$

1) Both statements I and II true.

2) Both statements I and II false.

3) Statement I is true but statement II is false.

4) Statement I is false but statement II is true.

Numerical Value Type:

12. $\frac{d}{dx}(\operatorname{cosec}^2 x - \cot^2 x) = \underline{\hspace{2cm}}.$

Matrix Match Type:13. **Column-I****Column-II**

a) The value of $\frac{d}{dx}(e^x)$ at $x = 1$ is

p) 10

b) $\frac{d}{dx}(10x) =$

q) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$

c) The value of $\frac{d}{dx}(5x^2 + 7)$ at $x = 1$ is

r) $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$

d) The value of $\frac{1}{2} \frac{d}{dx}(x^5 + 3x^3 + 6x)$ at $x = 1$ is

s) $\lim_{x \rightarrow \infty} \frac{10x^2 + 5}{x^2 + 3}$

JEE MAIN & ADVANCED**LEVEL - 2****Single Correct Choice Type:**

14. If $y = (\log x)^5$ then $\frac{dy}{dx} =$

1) $\frac{5(\log x)^4}{x}$

2) $5(\log x)^4$

3) $\frac{(\log x)^4}{x}$

4) $(\log x)^4$

15. $\frac{d}{dx}(\sqrt{\cos \sqrt{x}}) =$

1) $\frac{\cos \sqrt{x}}{4\sqrt{x} \sin \sqrt{x}}$

2) $\frac{\sin \sqrt{x}}{4\sqrt{x} \cos \sqrt{x}}$

3) $\frac{-\cos \sqrt{x}}{4\sqrt{x} \sin \sqrt{x}}$

4) $\frac{-\sin \sqrt{x}}{4\sqrt{x} \cos \sqrt{x}}$

16. If $y = a^{a^x}$, then $\frac{dy}{dx} =$

1) $y \cdot a^x (\log a)^2$

2) $y \cdot a^x \cdot \log a$

3) $(y \cdot a^x)^2$

4) $(y \cdot a^x)$

17. $\frac{d}{dx}[\log \{\log(\log x)\}] =$

1) $\frac{1}{x \log x \log \log(x)}$

2) $\frac{-1}{x \log x \log \log(x)}$

3) $\frac{x}{\log x \log(\log(x))}$

4) $\frac{1}{\log x \log(\log(x))}$

18. If $y = 2^{ax}$ and $\frac{dy}{dx} = \log 256$ at $x = 1$, then the value of a is
 1) 0 2) 1 3) 2 4) 3
19. $\frac{d}{dx} \left\{ e^{\log \sqrt{1+\cot^2 x}} \right\} =$
 1) $\operatorname{cosec} x \cdot \cot x$ 2) $-\operatorname{cosec} x \cdot \cot x$ 3) $\operatorname{cosec}^2 x \cdot \cot x$ 4) 0
20. $\frac{d}{dx} (\sin \sqrt{2x-3}) =$
 1) $\frac{\cos \sqrt{2x-3}}{\sqrt{2x-3}}$ 2) $\frac{-\cos \sqrt{2x-3}}{2\sqrt{2x-3}}$
 3) $\sqrt{2x-3} \cos \sqrt{2x-3}$ 4) $\cos \sqrt{2x-3}$
21. $\frac{d}{dx} \left\{ \log \left(x + \sqrt{a^2 + x^2} \right) \right\} =$
 1) $\frac{1}{(x + \sqrt{a^2 + x^2})}$ 2) $\frac{x}{\sqrt{a^2 + x^2}}$ 3) $\frac{1}{x(x + \sqrt{a^2 + x^2})}$ 4) $\frac{1}{\sqrt{a^2 + x^2}}$
22. If $f(x) = \frac{1}{(x+1)^5}$, then $f'(x) =$
 1) $-5 \cdot (x+1)^{-6}$ 2) $\frac{-5}{(x+1)^6}$ 3) $-5(x+1)^6$ 4) Both 1 & 2
23. If $f(x) = \frac{x^{100}}{100} + \frac{x^{99}}{99} + \dots + \frac{x^2}{2} + x + 1$, then $f'(1) =$
 1) 300 2) $100f'(0)$ 3) 200 4) $200f'(6)$

JEE MAIN & ADVANCED**LEVEL - 3****Single Correct Choice Type:**

24. If $f(x) = |x^2 - 5x + 6|$, then $f'(x) =$
 1) $2x - 5$ for $2 < x < 3$ 2) $5 - 2x$ for $2 < x < 3$
 3) $2x - 5$ for $2 \leq x \leq 3$ 4) $5 - 2x$ for $2 \leq x \leq 3$
25. If $f(0) = 0$, $f'(0) = 2$, then the derivative of $y = f(f(f(f(x))))$ at $x = 0$ is
 1) 2 2) 8 3) 16 4) 4

26. If $f(x) = \sqrt{1 + \cos^2(x^2)}$, then $f\left(\frac{\sqrt{\pi}}{2}\right) =$

1) $\frac{\sqrt{\pi}}{6}$

2) $-\sqrt{\frac{\pi}{6}}$

3) $\frac{1}{\sqrt{6}}$

4) $\frac{\pi}{\sqrt{6}}$

27. If $y = \left(x + \sqrt{x^2 + a^2}\right)^n$, then $\frac{dy}{dx} =$

1) $\frac{ny}{\sqrt{x^2 + a^2}}$

2) $\frac{-ny}{\sqrt{x^2 + a^2}}$

3) $\frac{nx}{\sqrt{x^2 + a^2}}$

4) $\frac{-nx}{\sqrt{x^2 + a^2}}$

28. If $y = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \dots \dots \frac{x^n}{n!}$, then $\frac{dy}{dx} =$

1) y

2) $y + \frac{x^n}{n!}$

3) $y - \frac{x^n}{n!}$

4) $y - 1 - \frac{x^n}{n!}$

SEEKING TO DESCRIBE

29. Find the derivative of following functions by using first principle

i) $e^x - \sec x - \cot x$

ii) $\cos x^3$

iii) $\cos(\sin x^2)$

SYNOPSIS 2

Product rule: The derivative of product of two functions is given by the following

$$\text{product rule } \frac{d[f(x) \cdot g(x)]}{dx} = \left[\frac{d}{dx}(f(x)) \right] \cdot g(x) + \left[\frac{d}{dx}g(x) \right] \cdot f(x)$$

Quotient Rule: The derivative of quotient of two functions is given by the following quotient rule (whenever the denominator is non - zero)

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \cdot \frac{d}{dx}(f(x)) - f(x) \cdot \frac{d}{dx}(g(x))}{(g(x))^2}$$

The above two rules may be restated in the following functions which aids in recalling them easily

let $f(x) = u$ and $g(x) = v$

$$(uv)' = u'v + uv'$$

This is referred to as Leibnitz rule for differentiating product of functions (or) the product rule.

Similarly the Quotient rule is $\left(\frac{u}{v} \right)' = \frac{v \cdot u' - u \cdot v'}{v^2}$

Example-1: The derivative of $f(x) = x^n$ can be computed in the following way

Let $f(x) = x^{n-1} \cdot x$

By product rule

$$\begin{aligned} \frac{d}{dx} f(x) &= \left[\frac{d}{dx}(x^{n-1}) \right] \cdot x + \left[\frac{d}{dx}(x) \right] \cdot x^{n-1} \\ &= (n-1) \cdot x^{n-2} \cdot x + 1 \cdot x^{n-1} \\ &= (n-1) \cdot x^{n-2+1} + x^{n-1} \\ &= x^{n-1} [n-1+1] \\ \frac{d}{dx}(x^n) &= nx^{n-1} \end{aligned}$$

Example-2: Find the derivative of $f(x) = \frac{x+1}{x}$

$$\begin{aligned} \frac{d}{dx}(f(x)) &= \frac{1}{x^2} \left[x \left(\frac{d}{dx}(x+1) \right) - (x+1) \cdot \frac{d}{dx}(x) \right] \\ &= \frac{1}{x^2} [x \cdot (1) - (x+1)1] = -\frac{1}{x^2} \end{aligned}$$

Parametric Differentiation: If $x = f(t)$ and $y = g(t)$ are the parametric equations of a curve then

$$(i) \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{g'(t)}{f'(t)}$$

$$(ii) \frac{d^n y}{dx^n} = \frac{d}{dt} \left(\frac{d^{n-1} y}{dx^{n-1}} \right) \left(\frac{dt}{dx} \right)$$

Example: If $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ then $\frac{dy}{dx}$

$$\frac{dx}{d\theta} = -3a \cos^2 \theta \cdot \sin \theta$$

$$\frac{dy}{d\theta} = 3a \sin^2 \theta \cdot \cos \theta$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3a \sin^2 \theta \cos \theta}{-3a \cos^2 \theta \sin \theta} = -\tan \theta$$

Derivative of Implicit Functions: If $f(x, y) = 0$, then differentiate each term w.r.t x reading y as a function of x and then collect the terms of $\frac{dy}{dx}$ together on left hand side and remaining terms on right hand side and then find $\frac{dy}{dx}$.

Another Method: If $f(x, y) = 0$ then $\frac{dy}{dx} = \frac{-\left(\frac{df}{dx}\right)}{\left(\frac{df}{dy}\right)} = \frac{\text{partial derivative of } f \text{ w.r to } x}{\text{partial derivative of } f \text{ w.r to } y}$

Example: If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$$a \cdot (2x) + 2h \left(x \cdot \frac{dy}{dx} + y \cdot (1) \right) + 2by \cdot \frac{dy}{dx} + 2g + 2f \cdot \frac{dy}{dx} = 0$$

$$2ax + 2hx \cdot \frac{dy}{dx} + 2hy + 2by \cdot \frac{dy}{dx} + 2g + 2f \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (2hx + 2by + 2f) = -(2ax + 2hy + 2g)$$

$$\frac{dy}{dx} (hx + by + f) = -(ax + hy + g)$$

$$\therefore \frac{dy}{dx} = \frac{-(ax + hy + g)}{hx + by + f}$$